

TEACHING PHILOSOPHY

I believe teaching is one of the most important and impactful professions in our society. It holds the power to shape minds, open possibilities, and influence an individual's life. Human progress, in many ways, reflects our ability to teach and learn from one another. It is through education that we share not only knowledge, but also the tools for critical thinking, creativity, and ethical reasoning.

Effective teaching is a deeply human centered process that can differ from one individual to the other. It requires adaptability, reflection and often a strong sense of empathy. I truly believe each student brings a unique set of experiences, strengths and learning styles to the classroom. What works for one might not work for the other. I aim to foster an inclusive environment where all students can feel supported, challenged and can thrive to their unique potential. My goal as an instructor/educator is to help create a space where students are encouraged to be independent thinkers, take intellectual risks and am committed to help them reach their full potential, both at the academic and personal level. I view teaching not just as a profession, but as a responsibility to contribute meaningfully to each student's journey. I am committed to adapt my approach to align with how each students learn best.

My teaching journey began right after high school when I started tutoring middle school students in mathematics and science. Since then, I've taught high school physics and mathematics, worked as a teaching assistant for both algebra and calculus based undergraduate physics courses including labs and discussions, graduate physics courses, and had the opportunity to guest lecture in multiple graduate-level courses. While teaching, I see my role not just as an instructor, but as a mentor and a guide that encourages significant student participation. This is accomplished through integrating personal response systems (Example: "clickers"), with interactive lecture demonstrations, think/pair/share activities, and whole-class discussions. Each group of students in my teaching voyage has been highly diverse. The willingness they show to collaborate and work together has eased promoting group discussion and sharing ideas to develop a strong conceptual understanding of physics principles and solve physics problems with proper analytical skills and reasoning. In my classroom, I focus on applying physics principles to real-world contexts which will help overcome misconceptions and relate to day-to day life, assign homework as a collaborative group task, and utilize computer-based-problem-solving techniques. Outside of the class, I hold office hours to help students with their homework, questions and confusion on either the course material or career goals. To summarize, my teaching philosophy implements the philosophy of Physics Education Research (PER) into classroom settings, recognizing that both learning is often a complex task and therefore I believe teaching can always evolve and improve.

I assess students via quizzes and exams that balance both conceptual understanding of physics with context-rich problem solving, online homework tutorials, and cooperative group work. In addition, I make use of rubrics that facilitates the use of criterion and evaluation tools; making it easier for the students to know precisely what is expected on each assignment and perform accordingly. Finally, to facilitate the impact between me and my students, I perform three stages of course feedback, one at the very start of the course, one midway and the final one at the end of

the course. This will help me understand the expectations of the students, monitor the progress and help them perform better and meet their goals.

In this journey so far, students have consistently shared that my clarity, approachability, and dedication have made a meaningful impact on their learning. These reaffirm the core of my teaching philosophy, that education is a relationship built on trust, clarity, and mutual respect. I strive to create a supportive and inclusive learning environment where students feel comfortable asking questions, exploring new ideas, and acknowledging when they're struggling. I am always open to improving the classroom experience experimenting with new methods that support student learning.

Ultimately, I take pride in the impact I've had on my students and remain committed to growing as an educator with every course I teach. Education is a shared journey, and I'm deeply grateful for the opportunity to walk this path along with my students.

Subash Adhikari
